

2. Roots of quadratic equations

What students need to learn:

Sum of roots and product of roots of a quadratic equation.

For the equation $ax^2 + bx + c = 0$
whose roots are α and β then $\alpha + \beta = -\frac{b}{a}$, $\alpha\beta = \frac{c}{a}$.

Manipulation of expressions involving the sum of roots and product of roots.

Knowledge of the identity $\alpha^3 + \beta^3 \equiv (\alpha + \beta)^3 - 3\alpha\beta(\alpha + \beta)$.

Forming quadratic equations with new roots.

For example, with roots $\alpha^3, \beta^3, \frac{1}{\alpha}, \frac{1}{\beta}, \frac{1}{\alpha^2}, \frac{1}{\beta^2}, \alpha + \frac{2}{\beta}, \beta + \frac{2}{\alpha}$; etc.
